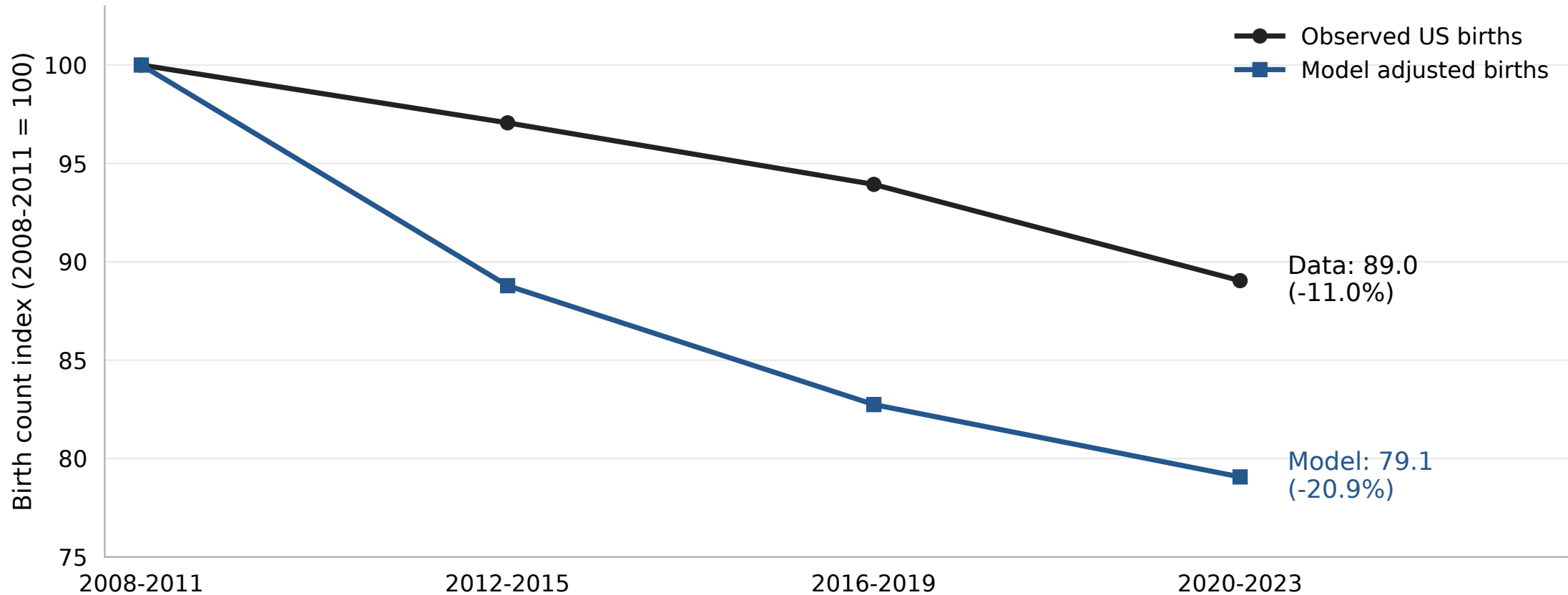


Aggregate births across four-year periods

Observed birth counts and the model's linear preference decline

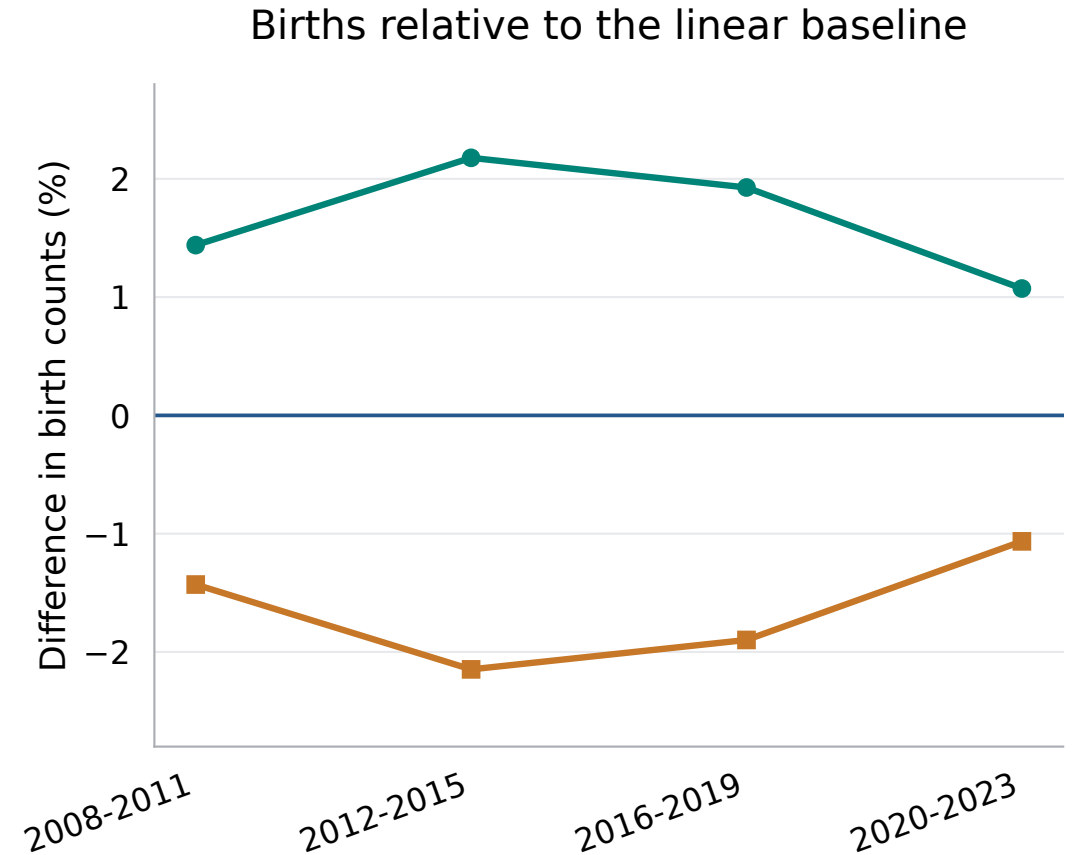
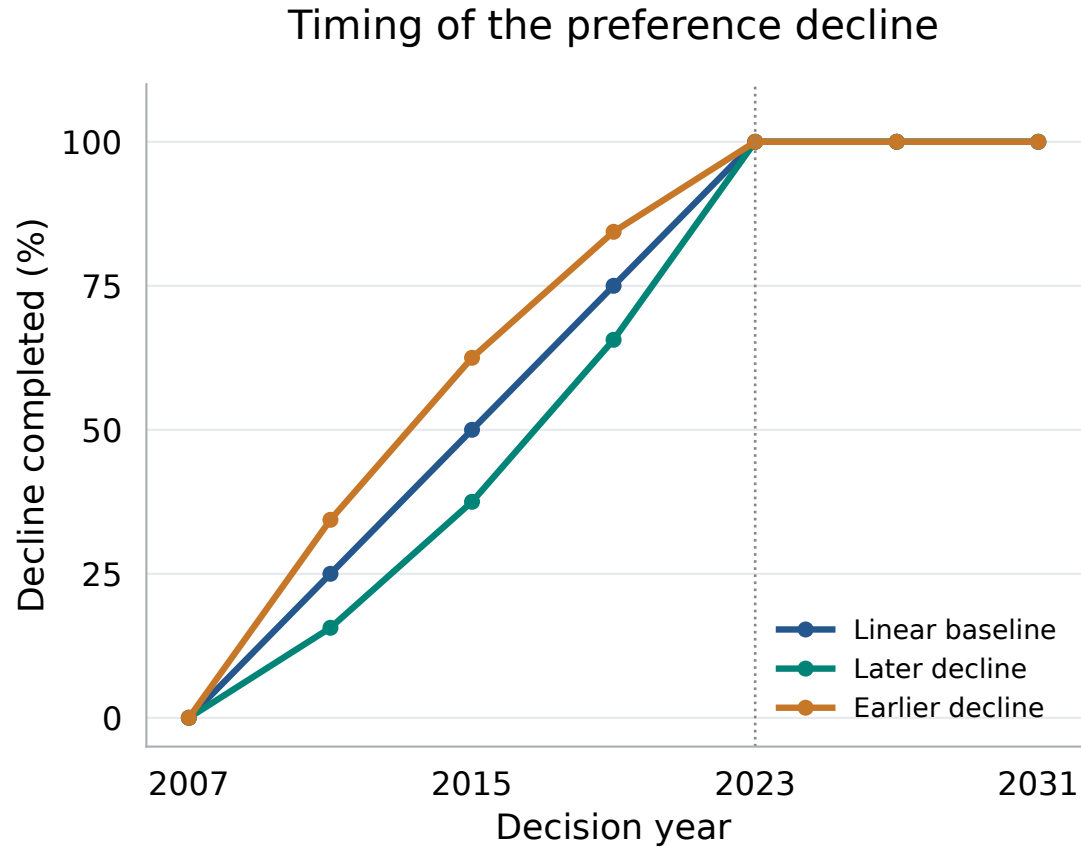


The 20.9% compares the last block with the first; it is not a cumulative shortfall.

$100 (B_{2020-2023}/B_{2008-2011} - 1) = -20.9\%$ for the model; -11.0% for the data.

Preference timing and the birth response

The full path is announced in 2007; preference endpoints and baseline prices are held fixed.

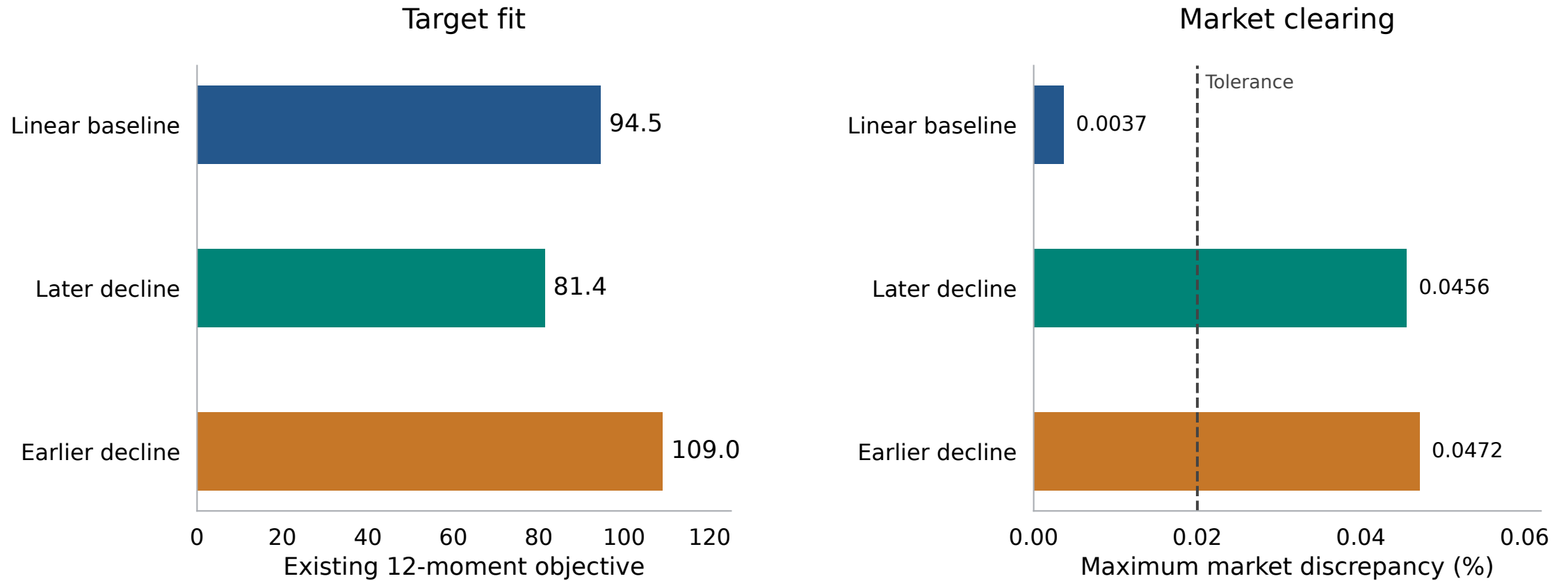


Timing shifts births by roughly 1-2% relative to the baseline, including the first block.

All three paths leave a first-to-last-block birth decline of about 21%, versus 11% in the data.

Conditional fit and housing-market clearing

Existing targets and weights; all structural parameters and baseline prices held fixed.



Later timing lowers the objective by 13.8%; both alternatives fail market clearing.

The birth-path gap improves only slightly; a full equilibrium comparison requires new prices.

Full target fit: linear baseline

Existing target system; parameters held fixed. All 12 rows are retained.

Moment	Target	Model	Gap	Weight	Contribution
Completed fertility (legacy tfr row)	1.918000	1.777103	-0.140897	1425.74	28.303715
Childlessness	0.188000	0.233597	0.045597	17180.7	35.719660
Mean age at first birth (years)	26.044627	26.233253	0.188626	44.4444	1.581326
Share of first births at age 30+	0.260327	0.244093	-0.016234	10000	2.635514
Rooms added after first birth	0.720246	0.411400	-0.308846	137.565	13.121816
Parent room gap: 3+ versus 1-2 children	0.367700	0.347411	-0.020288	2958.51	1.217789
Parent ownership gap	0.167662	0.146588	-0.021074	14229.6	6.319490
Ownership rate	0.575472	0.587508	0.012036	1207.85	0.174967
Mean occupied rooms, ages 18-85	5.779970	6.326208	0.546237	11.9732	3.572496
Wealth / annual gross labor earnings	6.873100	7.057293	0.184193	6.28767	0.213322
Annual bequest flow / aggregate wealth	0.008800	0.008354	-0.000446	5.16529e+06	1.025689
Old wealth/income: p90 / median	3.448111	3.346354	-0.101757	56.9598	0.589791

Total objective: 94.475576. Each contribution equals weight x gap squared.

Shares are fractions; housing outcomes are rooms. Measurement definitions remain those of the inherited target system.

These are conditional diagnostics. Terminal unit-rent distance is 1.088827% against a 1% tolerance for all three cases.

Birth-count figures include imputed additional births at entry into the 3+ child group; they do not measure female TFR.

Full target fit: later decline

Existing target system; parameters held fixed. All 12 rows are retained.

Moment	Target	Model	Gap	Weight	Contribution
Completed fertility (legacy tfr row)	1.918000	1.795799	-0.122201	1425.74	21.290581
Childlessness	0.188000	0.229399	0.041399	17180.7	29.446252
Mean age at first birth (years)	26.044627	26.239008	0.194380	44.4444	1.679277
Share of first births at age 30+	0.260327	0.244331	-0.015997	10000	2.558989
Rooms added after first birth	0.720246	0.411882	-0.308365	137.565	13.080916
Parent room gap: 3+ versus 1-2 children	0.367700	0.344747	-0.022953	2958.51	1.558605
Parent ownership gap	0.167662	0.146732	-0.020930	14229.6	6.233229
Ownership rate	0.575472	0.587417	0.011945	1207.85	0.172332
Mean occupied rooms, ages 18-85	5.779970	6.327499	0.547528	11.9732	3.589401
Wealth / annual gross labor earnings	6.873100	7.055364	0.182264	6.28767	0.208878
Annual bequest flow / aggregate wealth	0.008800	0.008357	-0.000443	5.16529e+06	1.015201
Old wealth/income: p90 / median	3.448111	3.346354	-0.101757	56.9598	0.589791

Total objective: 81.423452. Each contribution equals weight x gap squared.

Shares are fractions; housing outcomes are rooms. Measurement definitions remain those of the inherited target system.

These are conditional diagnostics. Terminal unit-rent distance is 1.088827% against a 1% tolerance for all three cases.

Birth-count figures include imputed additional births at entry into the 3+ child group; they do not measure female TFR.

Full target fit: earlier decline

Existing target system; parameters held fixed. All 12 rows are retained.

Moment	Target	Model	Gap	Weight	Contribution
Completed fertility (legacy tfr row)	1.918000	1.758610	-0.159390	1425.74	36.221260
Childlessness	0.188000	0.237762	0.049762	17180.7	42.544408
Mean age at first birth (years)	26.044627	26.227052	0.182424	44.4444	1.479050
Share of first births at age 30+	0.260327	0.243809	-0.016519	10000	2.728683
Rooms added after first birth	0.720246	0.410920	-0.309327	137.565	13.162648
Parent room gap: 3+ versus 1-2 children	0.367700	0.350076	-0.017623	2958.51	0.918852
Parent ownership gap	0.167662	0.146459	-0.021203	14229.6	6.396955
Ownership rate	0.575472	0.587607	0.012135	1207.85	0.177879
Mean occupied rooms, ages 18-85	5.779970	6.324939	0.544968	11.9732	3.555913
Wealth / annual gross labor earnings	6.873100	7.059240	0.186140	6.28767	0.217857
Annual bequest flow / aggregate wealth	0.008800	0.008352	-0.000448	5.16529e+06	1.036325
Old wealth/income: p90 / median	3.448111	3.346354	-0.101757	56.9598	0.589791

Total objective: 109.029620. Each contribution equals weight x gap squared.

Shares are fractions; housing outcomes are rooms. Measurement definitions remain those of the inherited target system.

These are conditional diagnostics. Terminal unit-rent distance is 1.088827% against a 1% tolerance for all three cases.

Birth-count figures include imputed additional births at entry into the 3+ child group; they do not measure female TFR.

Inherited parameters and restrictions

All 11 previously estimated coordinates are held fixed in the timing comparison.

Parameter	Value	Lower bound	Upper bound	Role	Near bound
beta_annual	0.995277	0.94	0.9995	Held fixed	No
kappa_fert	2.168173	0.02	50	Held fixed	No
kappa_fert_continuation	1.770771	0.02	50	Held fixed	No
chi	1.053727	0.1	5	Held fixed	No
H0	13.716049	0.2	80	Held fixed	No
theta0	0.570350	0	8	Held fixed	No
theta1	0.103724	0.02	16	Held fixed	Yes
hbar_child_rooms	0.247085	0.1	1.8	Held fixed	No
first_birth_fixed_cost	4.619731	0	8	Held fixed	No
hbar_first_child_jump	0.469765	0	0.5	Held fixed	No
psi_child_change_2023	-0.321388	-1.5	0.2	Held fixed	No
psi_child_2007	0.290052	-	-	Normalized	No
psi_child_2023	-0.031336	-	-	Derived	No
tenure_choice_kappa	0.005000	-	-	External	No
housing_supply_elasticity	0.630000	-	-	External	No

Bounds and near-bound flags reproduce the inherited search domain. No parameter estimate changed in this exercise.